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Effect of pretreated and anaerobically digested microalgae on the chemical and biochemical properties of soil and wheat grown on fluvisol

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ABSTRACT

In this study, the effects of the potential application of digestate as an agricultural fertiliser obtained from anaerobically digested microalgae treated by three pretreatment methods, namely alkaline hydrogen peroxide (AHP), high temperature and pressure (HTP), and hydrodynamic cavitation (HC) on some properties of soil, and wheat growth and yield were investigated. For this purpose, pretreated and anaerobically digested microalgae digestates alone or together with diammonium phosphate (DAP) as a chemical fertiliser were applied to soil for wheat growth. The highest dosage of AHP pretreated digestate combined with a half dose of DAP applied to soil was rich in nutrients as 0.25%N and 7.19 mg kg⁻¹ compared to all groups. The properties of the soils were enhanced by applying the highest dosage (0.06 g kg⁻¹) of microalgae digestate combined with a half dose of DAP. 0.02 g kg⁻¹ dosage of HC pretreated digestate combined with a half dose of DAP also greatly improved nitrogen use efficiency indices by up to 104%. The soils' enzyme activities increased in wheat growth experiments by applying either raw or pretreated microalgae digestates. The soils' β -glycosidase, alkaline phosphatase, and urease enzyme activities increased to 1.38 mg pNP g⁻¹ soil, 4.91 mg pNP g⁻¹ soil, and 2.27 mg NH₄-N 100 g⁻¹ soil respectively by the application of highest dosage of HC pretreated digestate. The digestates did not have a toxic effect on wheat growth, it was determined that applied pretreatment processes did not cause significant changes in wheat plant height or wet and dry weight.

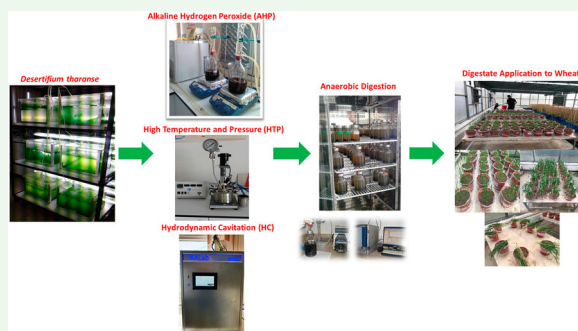
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1. Introduction

Anaerobic digestion applications have gained importance in the last 30 years due to the intense demands on alternative energy sources and their influential role in sustainable waste management. In recent years, microalgae have been subjected to extensive research as an alternative raw material for biofuel production [1–4]. Microalgae have some advantages over traditional energy sources, such as high growth rate, short

harvesting cycle, high photosynthetic efficiency, and high biomass yield [5]. There is increasing interest in using biological fertilisers to reduce the use of chemical fertilisers that cause serious environmental problems like water pollution and greenhouse gas emissions [6–9]. The soil is one of the most precious natural sources and all existence continues on it [10]. To sustain the increasing food demand, productivity should be increased without adversely affecting soil health [11]. This can be facilitated

by expanding the use of biofertilisers. Biofertiliser applications can increase the development of microorganisms in the soil, improving soil fertility over time [12]. Biofertilisers are cost-effective, environmentally friendly, and sustainable alternatives to conventional chemical fertilisers, because they not only increase agricultural production but also minimise the negative impact on land usage and water resources [13,14]. In this context, the use of algae biomass (macro and micro) has been widely researched as a fertiliser for agricultural production as a 'green economy' alternative product [15,16].

There is an increasing interest in microalgae, which has the potential for biofertilisation, biostimulants, bio-control agents properties for agricultural use [17–23]. Microalgae-sourced biofertilisers have the potential to meet the nutrients that plants need from the soil. They also contain trace elements and natural phytohormones that are necessary for the proper growth and development of plants [17,18]. Besides, using microalgae as bio-fertilisers offers benefits like using their N-rich biomass, carbon retention, improved soil health, water retention in the soil, stability of soil aggregates, and prevention of nutrient losses [24].

Anaerobic digestion (AD) process is used to convert organic wastes into methane, a medium calorific value gas. AD is described as an energy-efficient and cost-effective biotechnological process [25]. It is a widely applied, proven, and accepted technology that satisfies the requirements for both waste management and energy production [26]. Macronutrients such as nitrogen (N), phosphorus (P), and potassium (K), which plants need, remain in a solid–liquid substance called 'digestate' or 'fermented fertiliser' that is generated after biogas production by AD. In addition to its use as renewable energy, biogas also offers many advantages as a plant nutrient by separating solid–liquid after anaerobic digestion [27]. Digestate can be used as organic fertiliser or soil conditioner for its organic and mineral components [28]. According to previous studies, biogas digestates can potentially be used in crop nutrition to provide NPK nutrients and reduce the demand for chemical fertilisers [29,30].

Several studies [31–33] have pointed out the necessity of recycling nutrients through digestate reuse to improve the sustainability of biogas production and the circularity of bio-based fertilisers. Though, there is still incredibly limited research on how the digestate from microalgae and macroalgae affects the growth of cress [24], wheat [34], and tomatoes [24,35], or the fertiliser properties. Bio-based fertilisers support the circular economy approach and the European Commission has set a goal of a 30% reduction of non-renewable

resources in fertiliser production. Hence, microalgae and macroalgae, along with bio-based wastes, can be essential sources for the fertiliser industry [36].

Numerous physical, chemical, and biological pretreatment studies have been carried out to increase biogas production from macroalgae and microalgae. With the selected pretreatment method, disintegration-resistant cell walls can dissolve more easily and methane yield can be increased by accelerating microalgae hydrolysis. Several studies have aimed to maximise methane production with minimum energy input. Accordingly, the methane yield of microalgae residues ranges from 0 to 185% higher than raw microalgae [37–39].

Most studies have focused on the enhancement of methane production from macroalgae and microalgae [40–42]. Although some researchers have investigated the effect of raw digestate of macroalgae and microalgae from AD on soil properties [43], to the best of our knowledge, there is no study regarding the application of microalgae digestate pretreated by alkaline hydrogen peroxide (AHP), high temperature and pressure (HTP), and hydrodynamic cavitation (HC), and its effect on wheat-grown soil [44,45].

The research perspective of this manuscript includes the questions that need to be deeply investigated and the orphan topics. Although applying pretreated digestates from microalgae has been limited as fertilisers, their effects on the soil properties and crops vegetation are still unclear. Shedding light on this issue is awkward due to the diversity of microalgae for AD, the variety of operational parameters of AD, pretreatment type, conditions and effects, and the kind of plant grown. Therefore, this study aims to determine and compare the effects of microalgae digestates obtained from three pretreatment methods combined with anaerobic digestion on the chemical and biochemical properties of soil and wheat growth and yield. Correspondingly, this study focused on exploring digestate from cyanobacteria *Desertifilum tharense* with specific objectives of investigating: (1) application of three pretreatment methods as AHP, HTP, and HC to cyanobacteria (2) mineral compositions of pretreated and anaerobically digested cyanobacteria digestates (3) effects of pretreated and anaerobically digested cyanobacteria digestates on chemical and biochemical properties of soil (4) changes in enzyme activities (β glucosidase, Alkaline phosphatase, and Urease) of wheat-grown soil by applying pretreated cyanobacteria (5) effects of pretreated cyanobacteria digestate on the wheat yield components and elemental composition of the wheat plant (heights, wet and dry weights, elemental composition (N, P, and K).

2. Materials and methods

2.1. The isolation of microalgae

Cyanobacterial microalgae were isolated from the thermal Saraykoy spring in the city of Denizli, located in the southwest of Turkey, and were identified as '*Desertifilum tharense*' using 16S rRNA amplification with RT-PCR, DNA sequencing, and blast method with the NCBI online blast engine. Culture-specific parameters and compositional characterisation were described in Perendeci et al. [3]. Growth conditions and harvesting techniques of *D. tharense* can be found in Fardinpoor et al. [46]. *D. tharense* was subjected to a combination of pretreatment and anaerobic digestion to obtain digestate for further evaluation.

2.2. Pretreatment, biogas, and digestates

The digestates used in this study were obtained from different types of sequential pretreatment processes and anaerobic digestion of *D. tharense*. The *D. tharense* was subjected to three pretreatment processes, namely alkaline hydrogen peroxide (AHP), high temperature and pressure (HTP), and NaOH-assisted hydrodynamic cavitation (HC). All pretreatment processes were optimised and process variables were determined from the point of maximum methane production. For the experimental designs of these pretreatment methods, the central composite design (CCD) method was preferred. Table 1 shows the independent variables and the optimised pretreatment conditions for maximum methane production.

AHP pretreatment was performed in 1000 mL glass reactors. After adding the calculated quantity of deionised water to the reactor, pH was set to 11 with 10 M NaOH. Then, H₂O₂ was supplemented to the reactor and stirred for 1 minute [47]. The desired amount of

D. tharense was added and the pH was readjusted to 11.5. Finally, the reactor was closed with a condenser to prevent vapor–liquid forfeit and put in a preheated oil bath. The reaction time was initiated after the reactor contents reached the target temperature. During pretreatment, temperature was kept at the target level using a temperature controller and the reactors were continuously stirred with magnetic stirrers at 200 rpm.

HTP pretreatment was performed in a 300 mL stainless steel reactor (Parr series 5500 high-pressure reactor; Parr Instrument Company, USA) outfitted with a temperature controller and magnetically operated turbine agitator. The determined quantity of *D. tharense* was supplemented into the reactor and a reaction volume of 200 mL was set by adding deionised water. Then, the agitator was initiated and the reactor heated to the targeted working temperature. The reaction time was initiated when the reactor temperature reached the target level.

HC reactor with 1000 mL working volume for the pretreatment of *D. tharense* was used. The configuration includes a HC reactor, a pump, a flowmeter, a pressure gauge, an orifice, a cooling system, and a process control system. The internal pipe diameter and throat diameter of the orifice were selected as 10 mm and 3 mm, respectively. Detailed information can be found in Fardinpoor et al. [46]. The defined quantities of *D. tharense* (as 1.5% VS basis) and NaOH (w/w) were supplemented into the HC reactor and the content was passed through the bypass line for 5 minutes to achieve a homogeneous mixture. At the end of the reaction times of all pretreatment processes, the temperature of the contents of the reactors was rapidly cooled down to 30–35°C.

A biochemical methane potential (BMP) test was performed to determine the biochemical methane

Table 1. The ranges of selected independent variables and optimised pretreatment conditions for maximum methane production.

Alkaline hydrogen peroxide (AHP) pretreatment		
Reaction Temperature (°C)	40–100	Optimum Conditions: Reaction Temp.: 100°C Reaction Time: 1 h H ₂ O ₂ Conc. (%): 1.11 Cyanobacteria <i>D. tharens</i> e Conc. (%): 1.5
Reaction Time (h)	1–5	
H ₂ O ₂ Concentration (%)	0–3	
High temperature and pressure (HTP)		
Reaction Temperature (°C)	100–200	Optimum Conditions: Reaction Temp.: 100°C Reaction Time: 15 min. Cyanobacteria <i>D. tharens</i> e Conc. (%): 1.5
Reaction Time (Min.)	10–20	
NaOH assisted hydrodynamic cavitation (HC)		
Cavitation Number (Cv)	0.3–0.7	Optimum Conditions: Cavitation Number: 0.3 NaOH Conc. (%): 0 Reaction Temp: 30°C Cyanobacteria <i>D. tharens</i> e Conc. (%): 1.5 Reaction Time: 4 h
NaOH Concentration (%)	0–4	

potentials of pretreated microalgae. This method is based on the principle of incubating a certain amount of *D. tharensis* mixed with an anaerobic sludge at mesophilic conditions (35°C) and analysing biogas composition with the produced gas volume. The BMP test protocol according to Perendeci et al. [48] was applied.

2.3. Greenhouse experimental setup

Greenhouse experiment was conducted at the Department of Soil Science and Plant Nutrition in the Faculty of Agriculture at Ankara University. The experiment was performed on a dry weight basis, by placing 4 kg of soil in PVC pots (28 cm in top and 22.5 cm in bottom diameter) during the vegetation period (04.10.2018 – 18.12.2018) in a naturally lighted glasshouse. Experimental soil was collected from the plough layer (0–30 cm) of the experimental field of the Faculty of Agriculture at Ankara University (39°57'44.51" N; 32°51'46.95" E). We applied three different doses (0.01, 0.02, and 0.06 g kg⁻¹) of digestates (from raw *D. tharensis* and pretreated *D. tharensis* by AHP, HTP, and HC) to the soil alone. These doses were determined by considering the results obtained from the previous phase of the study (growing wheat in soil and soilless environments) and the examples in the literature. 150 kg ha⁻¹ DAP fertiliser was applied according to the farmer's experience (0.24 g pot⁻¹). We also applied a half dose of DAP fertiliser (75 kg ha⁻¹; 0.12 g pot⁻¹ DAP) in combination with digestates. The experimental plan included the control treatment that received no application.

Chemical fertilisers were mixed with soil before sowing and digestates were applied as a mixture of solid and liquid after sowing. Bread wheat (*Triticum aestivum* cv. Tosunbey) was used in the experiments. Twenty wheat seeds were planted in each pot at a depth of 6 cm; water was added to all pots at the volume required to reach 70% of the field capacity. The pots were closed in a way to provide ventilation and prevent sudden drying. When the germination of the plants started, the covers of the pots were removed entirely. Each pot was monitored individually from germination, with ten plants left in each pot. We also added an additional 50 mg kg⁻¹ N as NH₄NO₃ to meet the N requirement of wheat in the relevant pots 30 day after sowing. No pesticides were used. The wheat plant was harvested from the bottom closest to the soil at the end of a 10-week vegetation period from the establishment of the greenhouse experiment. Then, the plants were washed once with tap water and twice in deionised water. The plants were then dried at 65°C. Plant dry weights were measured and subsequently ground (200 µm sieve) for plant analyses.

2.4. Analysis of soil and plant samples

The soil samples were air-dried, ground, and sieved through a 12-mesh at the end of greenhouse experiment. The soil samples were analysed for pH, total nitrogen (N), available phosphorus (P) and potassium (K), urease, β-glycosidase, and alkaline phosphatase enzyme activity. The texture and available K of the soil samples were determined according to the U.S. Salinity Lab. Staff [49]. Total N was analysed by the Kjeldahl method [50] and available P was measured according to the procedure proposed by Olsen et al. [51]. pH was determined in 1:2.5 soil:water ratio (w/v) as proposed by Jackson [52]. β-glycosidase and alkaline phosphatase enzyme activities were measured spectrophotometrically [53] and urease activity was detected according to Hoffman and Teicher [54]. Total nitrogen (N) in the plant samples was determined by the Kjeldahl method. The plant samples were digested with a 4:1 acid mixture (HNO₃-HClO₄ 65 and 60% pure, Merck respectively) [55]. Total P and K concentrations were determined by ICP-OES (Perkin Elmer Optima 2100 DV, Waltham, MA, USA). All analyses were performed in triplicate. We also measured Partial Factor Productivity (PFP), a fertiliser-based Nitrogen Use Efficiency (NUE) index. PFP was calculated as crop yield per unit of applied nitrogen [56].

$$\begin{aligned} \text{Partial Factor Productivity(PFP)} \\ = \text{Yield/Fertilizer N} \end{aligned} \quad (1)$$

where PFP is the total N uptake by wheat, yield is the dry weight of wheat (g), and fertiliser N is the total amount of N added in the fertiliser treatment (mg kg⁻¹).

Another NUE index, Agronomic Efficiency (AE) was calculated as crop yield per unit of applied nutrient [56].

$$\begin{aligned} \text{The Agronomic Efficiency (AE)} \\ = (\text{Yield}_1 - \text{Yield}_0)/\text{Fertilizer N} \end{aligned} \quad (2)$$

where AE is the total N uptake by wheat, Yield₁ is the dry weight of control application (g), Yield₀ is the dry weight of other applications (g), and fertiliser N is the total amount of N added in the fertiliser treatment (mg kg⁻¹).

2.5. Statistical analyses

The greenhouse experiment was set up in a completely randomised design and each treatment contained three replicate pots. Analysis of variance (ANOVA) was performed on the data using MINITAB 17.1.0, and the significant differences between the treatment groups were calculated by Duncan's multiple range test ($p < 0.05$) using MSTAT-C.

3. Results and discussion

3.1. Mineral compositions of digestates

The mineral composition of digestates and their usability in agriculture were compared and evaluated by the relevant legislations [57,58]. Table 2 shows the mineral composition of digestates obtained from different pretreatment processes and anaerobic digestion of cyanobacteria *D. tharense*.

The digestate obtained from the combination of HTP and anaerobic digestion had a significantly lower organic matter (OM) content than the other digestates. The digestates obtained from HTP had lower OM than those obtained from AHP and HC. HTP pretreatment caused more OM solubilisation and more anaerobic digestion performance compared to AHP and HC. Previous research has compared pretreatment methods and explained the different solubilisation degrees related to the contents of the biomass [38,59,60]. While cavitation enhances methane production with a minor effect on solubilisation, thermal pretreatment has a high effect on solubilisation and methane production. In a general sense, the OM content of all digestates was in accordance with the limit values in the relevant regulations. The anaerobically digested samples pretreated with AHP, HTP, and HC had higher pH than the raw material. The digestate from raw *D. tharense* had a pH of 6.2; the pH levels of the anaerobically digested samples pretreated with AHP, HTP, and HC ranged from 7.34 to 7.77, indicating leakage of the cell content to the medium. The anaerobically digested samples pretreated with AHP, HTP, and HC had lower N contents than the digestate from *D. tharense*. The N contents of the pretreated samples varied based on the pretreatment. This may have resulted from more mineralisation in these samples. The N values of all digestates were consistent with the regulations on the product labels. According to Appels et al. [61] the optimal C to N ratio for anaerobic digestion is within the range of 25/30. However, in this

study, all digestates displayed lower C/N ratios. Phosphorus was found to be more available in digestates pretreated with HTP and HC compared to those pretreated with AHP. The anaerobically digested samples pretreated with AHP, HTP, and HC displayed similar heavy metal contents with the digestate from raw material. Heavy metal content is a limiting factor in using anaerobic digestates as agricultural inputs. The heavy metal contents of the digestates obtained from anaerobic digestion and pretreatment were below the limit values declared by the Ministry of Agriculture in Turkey and the EU limit values (Table 2). All digestates presented appropriate heavy metal contents for soil application.

3.2. Effects of digestates on chemical and biochemical properties of soil

The clay-rich soil used in our greenhouse experiment is classified as a Calcaric Fluvisol according to the FAO-UNESCO soil map of the world. The soil contained 1.3% OM, 0.80 g kg⁻¹ total nitrogen, 1.42 mg kg⁻¹ available P, 256.83 mg kg⁻¹ available K, and a pH of 8.4.

Table 3 shows the soil analysis results of greenhouse experiment conducted by growing wheat plants under controlled conditions to reveal the effect of digestates from microalgae pretreated by AHP, HTP, and HC on plant growth and soil properties.

Overall, the application of chemical fertiliser decreased pH slightly, and the application of digestates increased pH slightly. On the other hand, changes in pH were not statistically significant between the applications (Table 3). Applying digestate increased pH more than control and DAP alone, though not statistically significantly. Still, digestate applications at high doses may have an increasing effect on soil pH in long-term use [62].

Total nitrogen (N) was higher in the control soil compared to DAP alone. However, the N content of the soil was the highest at 0.25% with the application of 0.06 g kg⁻¹ digestate from AHP treatment along with a half

Table 2. Mineral compositions of digestates obtained from the different types of pretreatment processes and anaerobic digestion of Cyanobacteria *D. tharense*.

Analysis parameters	Unit	RM	AHP	HTP	HC	Turkey regulation limit values [57]	EU regulation limit values [58]
Organic Matter	%	72.00	76.00	54.00	72.00	At least 20%	15%
pH	w/v	6.20	7.77	7.34	7.73	–	–
Total N	%	3.25	3.02	2.80	2.97	Declared if it exceeds 1%	2.5
C/N	%	12.85	14.59	11.18	14.06	–	–
Total P	%	0.38	0.31	0.37	0.39	Declared if it exceeds 1%	2
Total Cd	mg kg ⁻¹	<0.005	<0.005	<0.005	<0.005	3	1.5
Total Pb	mg kg ⁻¹	38.50	34.70	39.10	64.10	150	120
Total Cr	mg kg ⁻¹	60.50	51.40	61.70	61.50	350	2
Total Ni	mg kg ⁻¹	110.60	108.50	115.30	114.50	120	50
Total Cu	mg kg ⁻¹	238	206.90	256	218.02	450	200

RM = Raw material, AHP = Alkaline hydrogen peroxide, HTP = High pressure and temperature, HC = Hydrodynamic cavitation.

Table 3. Chemical properties of soils after the application of digestates.

Treatments	pH (1:2.5 w/v)	N (%)	Available P (mg kg ⁻¹)	Available K (mg kg ⁻¹)
Digestate from raw <i>D.tharanse</i>				
A	8.03	0.08d	3.26b	310.77cd
B	8.04	0.21a	6.65a	331.73c
C	8.04	0.09 d	3.32b	329.15c
D	8.04	0.09d	3.28b	342.15bc
E	8.03	0.11cd	3.25b	354.58b
F	8.04	0.15bc	6.95a	371.68ab
G	8.03	0.19ab	7.19a	392.60a
H	8.04	0.20ab	7.05a	365.22ab
LSD $p<0.05$	0.09514	0.04926	0.5329	26.82
Digestate from alkaline hydrogen peroxide (AHP) Pretreatment				
A	8.03	0.08d	3.26b	310.77bc
B	8.04	0.21b	6.65ab	331.73b
C	8.05	0.12c	3.23b	364.73b
D	8.05	0.11c	3.32b	361.20a
E	8.08	0.12c	3.44b	362.64b
F	8.10	0.19b	6.62ab	364.73b
G	8.10	0.18b	7.19a	370.97b
H	8.12	0.25a	7.19a	377.88b
LSD $p<0.05$	0.5264	0.02943	0.5004	21.83
Digestate from high temperature and pressure (HTP) pretreatment				
A	8.03	0.08d	3.26b	310.77d
B	8.04	0.21a	6.65ab	331.73c
C	8.04	0.11c	3.09b	382.08a
D	8.04	0.11c	3.07b	372.27b
E	8.05	0.13c	3.19b	390.50a
F	8.08	0.20ab	6.43a	355.06bc
G	8.08	0.20ab	7.46a	382.50a
H	8.08	0.18b	7.69a	367.09b
LSD $p<0.05$	0.4426	0.02322	1.077	17.80
Digestate from hydrodynamic cavitation (HC) pretreatment				
A	8.03	0.08d	3.26b	310.77c
B	8.04	0.21a	6.65a	331.73c
C	8.07	0.11c	3.22b	354.99b
D	8.07	0.11c	3.24b	356.25b
E	8.10	0.11c	3.81b	372.30ab
F	8.08	0.17b	7.14a	374.36ab
G	8.08	0.19ab	7.06a	363.16ab
H	8.10	0.19ab	7.17a	384.34a
LSD $p<0.05$	0.2625	0.02270	0.7390	22.21

A = Control, B = Chemical Fertilisation (150 kg ha⁻¹ DAP), C = Digestate 1. Dose (0.01 g kg⁻¹), D = Digestate 2. Dose (0.02 g kg⁻¹), E = Digestate 3. Dose (0.06 g kg⁻¹), F = Digestate 1. Dose (0.01 g kg⁻¹) + Chemical Fertilisation (75 kg ha⁻¹ DAP), G = Digestate 2. Dose (0.02 g kg⁻¹) + Chemical Fertilisation (75 kg ha⁻¹ DAP), H = Digestate 3. Dose (0.06 g kg⁻¹) + Chemical Fertilisation (75 kg ha⁻¹ DAP). Lowercase letters represent significant differences among different treatments ($p<0.05$); data are means of three replicates.

dose of DAP chemical fertiliser. In the soils where digestates were applied together with DAP chemical fertilisers, N content was greater than applying digestates alone ($p<0.05$). Applying pretreated digestate alone increased the nitrogen content of the soils compared to the control soil, while applying the raw material alone (with a much higher N content (3.25%)) yielded similar results with the control soil (except for high dose). This may be because the pretreated digestate enhanced the solubilisation. According to Maria Solé-Bundó et al. [24] untreated, thermal pretreated, or co-digested digestates present suitable OM and macronutrients, especially organic and ammonium nitrogen in agricultural soils. They [24] reported that microalgae

digestates may contribute to the N content in the soil, but should be considered as soil conditioners rather than direct biofertilisers because they contain a moderate level of NH₄⁺-N.

The highest available P-value was found in soils that were applied DAP alone or together with digestates. Applying a half dose of DAP together with digestates increased the available P contents of soils more than DAP alone, which indicates that microalgae digestates are effective in increasing the P content of the soil, albeit to some extent (Table 3). The highest P content was found in the 'G' dosage of digestates from raw *D. tharense* and AHP, as well as from the 'H' dosage of digestates from HTP and HC. The effects of the digestates on the available P contents were statistically significant ($p<0.05$). Due to factors such as adsorption, precipitation, or conversion to organic form in soils [63], only a very small fraction of the total P is available for plants in the form of orthophosphate or easily mineralised organic P. Biomass is expected to increase organic P mineralisation by soil microorganisms and thus the desorption of orthophosphate anions (HPO₄²⁻ and H₂PO₄⁻) into the soil solution [64]. Phosphorus is the second most crucial plant nutrient involved in plant growth. The average P content in the soil is about 0.05% (w/w), only 0.01% of which can be present for plants due to insoluble forms [65]. The amount of fixed phosphorus decreases with the increase of OM in the soil. The availability of phosphorus increases with the addition of organic matter [66]. During the decomposition of OM, a large amount of CO₂ is formed in the soil. Carbonic acid is formed from the dissolving of carbon dioxide in water, which plays a significant role in releasing P in calcareous soils. Research reports that the degradation products of OM turn the P in the soil into easily usable phospho-humic compounds in calcareous-alkaline soils. Thus, phosphate ions are released by the displacement of humate and phosphate anions, increasing P availability [67].

The digestates from raw *D. tharense* and those from AHP, HTP, and HC increased the exchangeable K content of the soil compared to the control soil and DAP alone (Table 3). Using digestate alone increased K more than DAP alone ($p<0.05$). Comparing each application individually, each pretreatment alone increased the K content of the soil more than the raw material. This may be because untreated microalgae is more stable than thermal pretreated digestate.

The critical factors in the desorption of potassium from the soil are the microbial population capable of dissolving K in the rhizosphere and the available K uptake by plant species [68]. In this study, the amount of

available K was increased by the 'G' dosage of digestates from raw *D. tharense* and HTP, and by the 'H' dosage of digestates from AHP and HC.

3.3. Changes in soil enzyme activities (β glucosidase, alkaline phosphatase, and urease) in greenhouse experiments

Soil enzymes can be used as an indicator and used as a sensitive method in determining the effects of environmental pollutants, agricultural practices, ecological differences, vegetation type, and different soil properties [69,70]. In each cultivated soil, there is a certain enzyme level according to that soil, and the amount and types of enzymes vary depending on the quality and amount of the harvest residues left in the soil and the type and amount of organic and inorganic fertilisers given [71–73]. Table 4 shows the changes in soil enzymes (β -glucosidase linked to the C cycle [β G], alkaline phosphatase linked to the P cycle [ALP], and urease (linked to the N cycle [UR]).

β -glucosidase was higher in digestates pretreated with AHP, HTP, and HC at high doses. Primarily, applying a high amount (0.06 g kg^{-1}) of digestate alone or together with 75 kg ha^{-1} DAP resulted in the highest β G in all treatments. The high β G indicates a high energy source for microorganisms and the presence of easily degradable OM in the environment.

All treatments significantly increased the ALP activity of the soil compared to the control soil. Hence, all digestate additions had a slightly more enhancing effect on ALP activity compared to adding fertiliser alone (Table 4). Alkaline phosphatase is closely related to OM content and is mostly associated with microbial C [74,75]. Therefore, it is considered the best way to determine the relative and total mass of the microbial population in the soil. While N fertilisation increases ALP activity, adding P fertiliser generally decreases ALP activity [76,77]. Olander and Vitousek [78] have stated that ALP activity decreases in the case of N deficiency in the soil and that applying N fertiliser to the soil has a positive effect on ALP activity. Therefore, ALP activity may decrease in chemical fertiliser applications containing P. In the current study, ALP activity was lower when applying DAP (Sample B) and higher when applying microalgae digestates without DAP (Table 4).

All microalgae digestates were effective in increasing UR activity compared to the control soil. There were significant differences between all microalgae digestates and control and DAP applications. As the dose of digestate pretreated with AHP, HTP, and HC increased (C, D and E treatments), UR activity increased significantly

Table 4. β -glucosidase, alkaline phosphatase and urease enzyme activities of wheat experiment soils.

Treatments	β -glucosidase (mg pNP g ⁻¹ Soil)	Alkaline phosphatase (mg pNP g ⁻¹ Soil)	Urease (mg NH ₄ ⁺ -N 100 g ⁻¹ Soil)
Digestate from raw <i>D.tharanse</i>			
A	1.10e	4.54c	0.83e
B	1.15d	4.80b	1.60c
C	1.10e	4.86a	2.06a
D	1.19c	4.89a	1.83b
E	1.28a	4.96a	2.07a
F	1.23b	4.84ab	1.80b
G	1.17cd	4.73b	0.97f
H	1.26ab	4.89a	1.29d
LSD $p<0.05$	0.03374	0.1490	0.1479
Digestate from alkaline hydrogen peroxide (AHP) pretreatment			
A	1.10c	4.54c	0.83e
B	1.15bc	4.80ab	1.60c
C	1.19b	4.90a	1.31d
D	1.11c	4.88ab	1.39d
E	1.26a	4.98a	1.80c
F	1.096c	4.87ab	3.46a
G	1.14bc	4.80b	2.95b
H	1.19b	4.80b	2.88b
LSD $p<0.05$	0.04573	0.1310	0.2316
Digestate from high temperature and pressure (HTP) pretreatment			
A	1.10d	4.54b	0.83ge
B	1.15c	4.80ab	1.60d
C	1.18c	4.90a	2.30b
D	1.19c	4.84ab	2.81a
E	1.39ab	4.92a	2.93a
F	1.24b	4.84ab	1.78c
G	1.39ab	4.81ab	1.50d
H	1.44 a	4.84ab	1.83c
LSD $p<0.05$	0.06018	0.1242	0.1687
Digestate from hydrodynamic cavitation (HC) pretreatment			
A	1.10d	4.54c	0.83d
B	1.15d	4.80ab	1.60bc
C	1.22c	4.86a	1.53c
D	1.22c	4.87a	1.57c
E	1.38a	4.91a	2.27a
F	1.22c	4.88a	1.55c
G	1.30b	4.81a	1.56c
H	1.36a	4.89a	1.80b
LSD $p<0.05$	0.05662	0.1239	0.2217

A = Control, B = Chemical Fertilisation (150 kg ha^{-1} DAP), C = Digestate 1. Dose (0.01 g kg^{-1}), D = Digestate 2. Dose (0.02 g kg^{-1}), E = Digestate 3. Dose (0.06 g kg^{-1}), F = Digestate 1. Dose (0.01 g kg^{-1}) + Chemical Fertilisation (75 kg ha^{-1} DAP), G = Digestate 2. Dose (0.02 g kg^{-1}) + Chemical Fertilisation (75 kg ha^{-1} DAP), H = Digestate 3. Dose (0.06 g kg^{-1}) + Chemical Fertilisation (75 kg ha^{-1} DAP). Lowercase letters represent significant differences among different treatments ($p<0.05$); data are means of three replicates.

($p<0.05$; Table 4). Adding organic materials to the soil increases the microorganisms in the soil and its UR activity [79,80].

Soil enzyme activities were positively affected by the use of microalgae as biofertilisers [73,81,82]. Rao and Burns [83] have reported that microalgae fertilisation significantly enhances the dehydrogenase, UR, and ALP activities, resulting in higher OM mineralisation. De Caire et al. [81] also report that *T. tenuis* and *Microchaete tenera* inoculants in the soil increase the activity of extracellular enzymes like glucosidase, phosphomonoesterase, arylsulfatase, protease, and UR, as well as the accumulation of intracellular dehydrogenase.

Table 5. Heights, wet and dry weights, elemental composition (N, P, and K) of wheat, PFP, and AE.

Treatments	Plant height (cm)	Wet weight (g)	Dry weight (g)	N (%)	PFP (%)	AE (%)	P (%)	K (%)
Digestate from raw <i>D.tharanse</i>								
A	50.00b	13.60d	2.51b	2.06bc	–	–	0.15c	2.83c
B	52.66ab	18.21a	2.89a	2.93a	0.048b	0.034b	0.19ab	4.19ab
C	55.00 a	15.70bcd	2.69ab	1.88c	0.089a	0.088a	0.22a	4.85ab
D	55.66a	16.85abc	2.80ab	1.99bc	0.093a	0.090a	0.20ab	4.63ab
E	56.00a	15.10cd	2.79ab	2.04bc	0.089a	0.086a	0.17bc	4.78ab
F	56.33a	18.10ab	2.90a	2.21bc	0.096a	0.093a	0.23a	5.19a
G	53.66ab	17.89ab	2.85a	2.42b	0.095a	0.091a	0.21a	4.39ab
H	56.33a	18.42a	2.87a	2.27bc	0.095a	0.092a	0.20ab	3.63bc
LSD $p<0.05$	4.090	2.260	0.2748	0.4400	0.009431	0.01981	0.03631	1.149
Digestate from alkaline hydrogen peroxide (AHP) pretreatment								
A	50.00b	13.60b	2.51b	2.06cd	–	–	0.15d	2.83c
B	52.66ab	18.21a	2.89a	2.93a	0.048c	0.034b	0.19c	4.33abc
C	50.00b	14.05b	2.56b	1.65e	0.085ab	0.084a	0.19c	5.59a
D	50.33b	12.41b	2.59b	1.77de	0.076b	0.073a	0.17cd	4.90ab
E	55.33a	14.25b	2.59b	1.69e	0.083ab	0.081a	0.19c	3.87bc
F	56.00a	19.09a	2.86a	2.56b	0.095a	0.092a	0.23b	4.65ab
G	56.00a	18.93a	2.84a	2.38bc	0.091a	0.088a	0.27a	5.46ab
H	54.66a	17.14a	2.87a	2.43b	0.085ab	0.082a	0.24ab	4.19abc
LSD $p<0.05$	4.298	2.197	0.2783	0.3393	0.009592	0.01997	0.03622	1.470
Digestate from high temperature and pressure (HTP) pretreatment								
A	50.00b	13.60c	2.51d	2.06c	–	–	0.15e	2.83b
B	52.66ab	18.21b	2.89bc	2.93a	0.048e	0.034b	0.19bcd	4.19ab
C	53.00ab	12.99c	2.49d	2.12bc	0.076d	0.075a	0.18cde	4.80a
D	55.00ab	14.20c	2.60d	1.99c	0.083c	0.080a	0.17de	4.57a
E	52.66ab	13.61c	2.91b	2.00c	0.083c	0.080a	0.18cde	4.68a
F	57.66a	18.45ab	2.82c	2.32bc	0.094b	0.091a	0.21abc	5.06a
G	57.00a	19.99ab	3.09a	2.30bc	0.103a	0.099a	0.22ab	4.56a
H	55.00ab	20.59a	3.04ab	2.42b	0.101a	0.098a	0.23a	4.39a
LSD $p<0.05$	5.966	2.068	0.1740	0.3061	0.005538	0.01820	0.03144	1.374
Digestate from hydrodynamic cavitation (HC) pretreatment								
A	50.00c	13.60c	2.51c	2.06cd	–	–	0.15d	2.83b
B	52.66bc	18.21b	2.89b	2.93a	0.048d	0.034c	0.19bc	4.19ab
C	54.33b	14.86c	2.49c	2.01cd	0.083c	0.081ab	0.16cd	4.11ab
D	55.66cd	14.87c	2.61c	1.98d	0.087c	0.084ab	0.17cd	4.39ab
E	52.33bc	14.23c	2.89b	1.96d	0.083c	0.080b	0.16d	4.46a
F	56.00b	20.04ab	2.88b	2.31bc	0.096b	0.093ab	0.23a	3.70ab
G	61.00a	20.68a	3.14a	2.41b	0.104a	0.101a	0.23a	3.66ab
H	55.33b	15.61c	2.99b	2.27bc	0.083c	0.079b	0.20b	4.68a
LSD $p<0.05$	2.7820	2.0610	0.2241	0.2979	0.0074	0.0189	0.0279	1.4130

A = Control, B = Chemical Fertilisation (150 kg ha⁻¹ DAP), C = Digestate 1. Dose (0.01 g kg⁻¹), D = Digestate 2. Dose (0.02 g kg⁻¹), E = Digestate 3. Dose (0.06 g kg⁻¹), F = Digestate 1. Dose (0.01 g kg⁻¹) + Chemical Fertilisation (75 kg ha⁻¹ DAP), G = Digestate 2. Dose (0.02 g kg⁻¹) + Chemical Fertilisation (75 kg ha⁻¹ DAP), H = Digestate 3. Dose (0.06 g kg⁻¹) + Chemical Fertilisation (75 kg ha⁻¹ DAP). Lowercase letters represent significant differences among different treatments ($p<0.05$); data are means of three replicates.

3.4. The effects of digestate on the yield components and elemental composition of the wheat plant

Table 5 shows the heights of the wheat plants, their wet and dry weights, some macro (N, P, and K) and microelements, and some NUE indices (Partial-factor Productivity/PFP, and Agronomic Efficiency/AE).

Plant height measurements determined after harvesting the wheat plants were higher in raw digestate applications compared to control and DAP alone. Furthermore, plant heights were greater when applying microalgae digestate pretreated with HTP and HC, except AHP. On the other hand, wheat plant heights were higher when microalgae digestates were applied together with a half dose of DAP fertiliser. The longest wheat height (61 cm) was obtained from applying 0.02 g kg⁻¹ microalgae digestate pretreated with HC and

75 kg ha⁻¹ DAP. Statistically, wheat plant height generally increased significantly by applying increasing doses of digestate and DAP fertiliser ($p<0.05$) (Table 5). Figure 1 shows the effects of applying microalgae digestate as raw and pretreated with AHP, HTP, and HC on wheat plant height.

Wheat plant weight was higher when applying raw microalgae digestates of different doses compared to the control soil, but lower than DAP applications. On the other hand, wheat plant weight was close to the control soil when applying microalgae digestates pretreated with AHP, HTP, and HC pretreated, but still lower than DAP. In soils where pretreated microalgae digestates were applied together with a half dose of DAP, the biomass weights of the wheat plants increased significantly ($p<0.05$) compared to pretreated microalgae digestate applications. The highest wet and dry

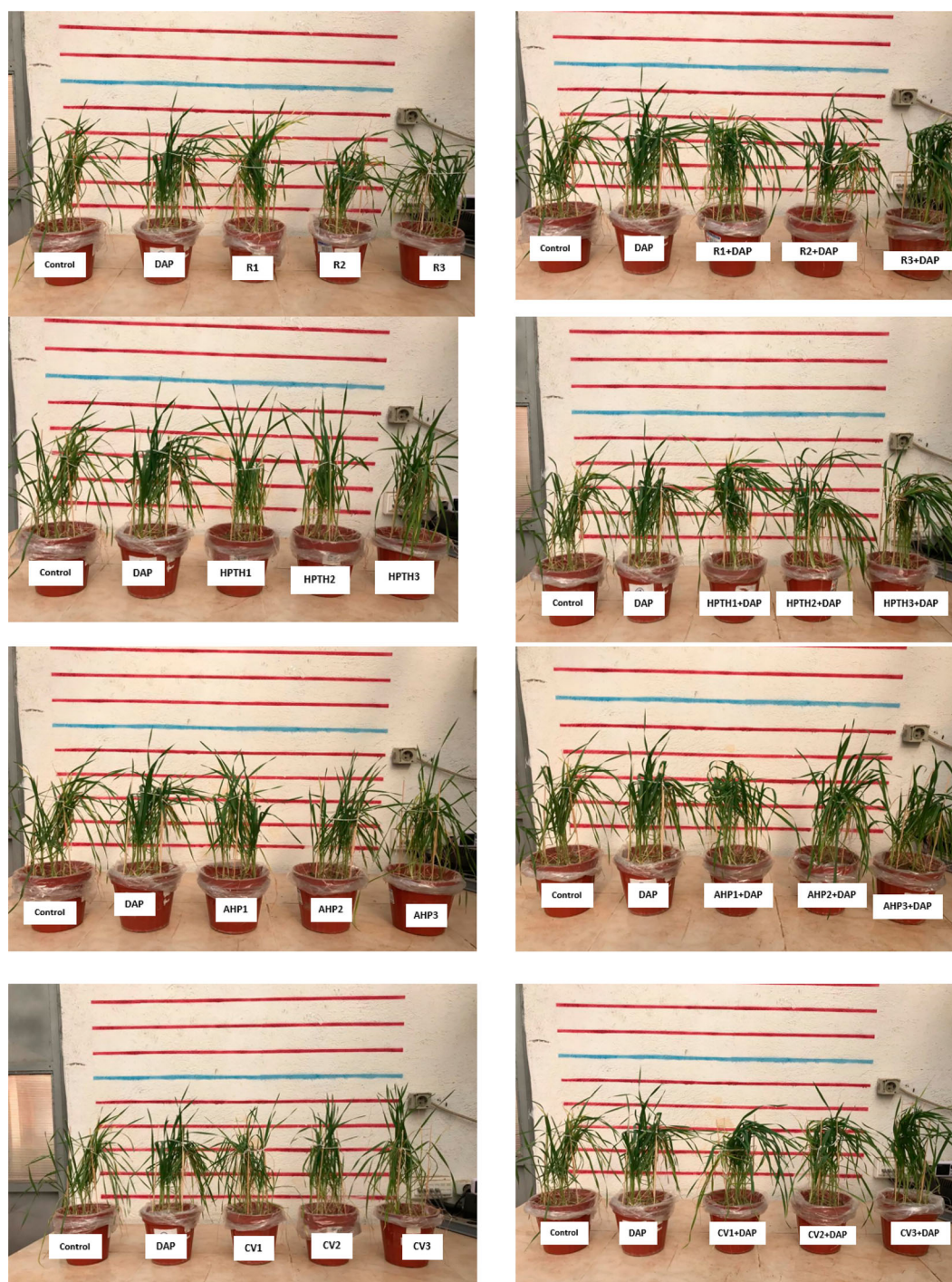


Figure 1. The effects of the application of microalgae digestate as raw and pretreated by AHP, HTP, and HC on wheat plant heights. R = Raw *D.tharanse*; DAP = chemical fertiliser as Diammonium Phosphate; AHP = Alkaline Hydrogen Peroxide; HTP = High Temperature and Pressure; HC = NaOH Assisted Hydrodynamic Cavitation.

biomass weight was measured as 20.68 g when applying 0.02 g kg^{-1} microalgae digestate pretreated with HC and 75 kg ha^{-1} DAP, similar to plant height (Table 5).

The highest dose of microalgae digestates alone or pretreated, 0.06 g kg^{-1} , was as effective as DAP fertiliser on plant height and biomass weights, which are indicators of plant growth. On the other hand, applying

microalgae digestates with a half dose of DAP was more effective on wheat plant growth. Applying microalgae digestate pretreated with AHP alone or together with DAP was not as effective as applying DAP alone on wheat plant height or biomass weight. The differences between the applications were statistically significant ($p < 0.05$) (Table 5). Previous studies report that

cyanobacteria applications to soils increase plant yield and these organisms raise the hormones stimulating plant growth and plant nutrients in the soil [84–88]. Numerous studies have highlighted the direct inoculation of cyanobacteria to the soil for various crops, explaining positive effects on germination plant growth and yield [89–93]. Research also reports that applying direct microalgae fertilisers can increase wheat yield, shoot/root length, and dry weight [94–96]. On the other hand, there is limited data on the effects of applying pretreated macroalgae and microalgae digestates on the fertiliser potential of agricultural products like tomatoes [45,48,97], *Lepidium sativum* [24], switchgrass forage [98], and wheat [34]. Elalami et al. [34] co-digested pretreated macroalgae (alkaline: KOH-5%, acid: H_3PO_4 – 5% at 70°C, and thermal: 70°C) and waste activated sludge. The effect of pretreated macroalgae digestates on germination index and dry weight were not influenced by the changes in pretreated macroalgae digestates in the control group [48].

The highest N content was found when applying DAP alone, while the lowest N content was found when applying microalgae digestate pretreated with AHP (Table 5). When the digestates were applied together with a half dose of DAP, the N contents were higher than applying digestate alone. Therefore, the digestates were not as effective as DAP fertilisers on the N content of wheat plants. The differences between the applications were statistically significant ($p < 0.05$). The N contents of the wheat plants were just below the sufficient amount in plants applied microalgae digestates alone, but adequate when the digestate was used together with a half dose of DAP.

NUE indices provide information on increasing crop yields while minimising inputs. PFP and AE are important performance indicators for NUE [43]. PFP shows how productive the current crop production system is relative to nutrient input [40,99], and AE is used to show how fertiliser input increases agricultural yield [43]. Contrary to the results regarding N content in the wheat plant, the lowest PFP and AE values were determined when applying DAP alone (Table 5). When digestates were applied alone or together with DAP, NUE indices (PFP and AE) were higher than DAP alone. The efficiency of using nitrogen fertiliser changes depending on the losses and the yield. The ability of the plant to absorb nitrogen, and the breakdown and binding effects of nitrogen affect the uptake and use of nitrogen by the plant [100]. Researchers report that the efficiency of nitrogen use would be lower at high nitrogen doses, improving NUE and reducing nitrogen loss by washing [101,102]. Our findings were similar in terms of NUE indices (PFP and AE).

Phosphorus contents of the wheat plants were higher in all applications compared to the control. Applying digestates with different pretreatments yielded similar P contents with applying DAP, while applying digestates together with a half dose of DAP significantly increased the P content (Table 5). The increase in P content in all applications due to rising microalgae doses and adding DAP was statistically significant compared to the control soils ($p < 0.05$). Applying digestates alone or together with a half dose of DAP increased the amount of K in wheat plants compared to the control group. Applying digestates pretreated with AHP and HTP were more effective in this regard, especially together with a half dose of DAP.

We determined that applying raw and pretreated microalgae digestates to the soil were influential on the amounts of P and K rather than N. Plant growth parameters are strongly correlated with soil fertility and any improvement in soil fertility can affect plant growth. Other studies have shown that various microalgal extracts, attributed to the presence of plant growth regulators such as auxins, gibberellins, or cytokines, and high rates of macro and micronutrients, increase the growth of agricultural crops [103].

Raw and pretreated microalgae digestates could be an alternative to chemical fertilisers in terms of wheat plant growth parameters. Furthermore, applying raw and pretreated microalgae digestates in combination with chemical fertilisers has significant effects on soil properties, elemental content, and biomass development.

4. Conclusion

Food production and human health should be improved together by cultivating ‘cleaner and healthier’ products using biofertilisers obtained from different sources together with minimum doses of chemical fertilisers and by protecting the environment and the limited natural resources. In the current study, the chemical and biochemical properties of the soils were enhanced by applying the highest dosage of microalgae digestates combined with a half dose of DAP. The highest plant height, wet weight, and dry weight were obtained by applying anaerobically digested microalgae digestates pretreated with HC and adding DAP fertiliser. Microalgae digestates, either raw or pretreated, increased soil enzymes, which improved the soil properties. Microalgae digestates also greatly improved NUE indices (PPF and AE). The digestate applications were influential on the phosphorus and potassium contents of the wheat plant rather than its nitrogen content. This research revealed that it is possible to reduce the amount of

chemical fertiliser usage in the soil by half with the application of digestates from anaerobically digested microalgae. Based on our findings, digestates obtained from microalgae have potential benefits as an agricultural biofertiliser. Promising results from this study, including the effect of pretreated and anaerobically digested microalgae on plants, need to conduct further research in greenhouse and field conditions by considering different climatic conditions, soil types, and plants.

Disclosure statement

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